



Improving ionic conductivity of garnet solid-state electrolytes using Gradient boosting regression optimized machine learning

Yue Ma^{a,b}, Shaoxiong Han^{a,b}, Yan Sun^{a,b}, Zhenming Cui^{a,b}, Pengyu Liu^{a,b}, Xiaomin Wang^a, Yongzhen Wang^{a,b,*}

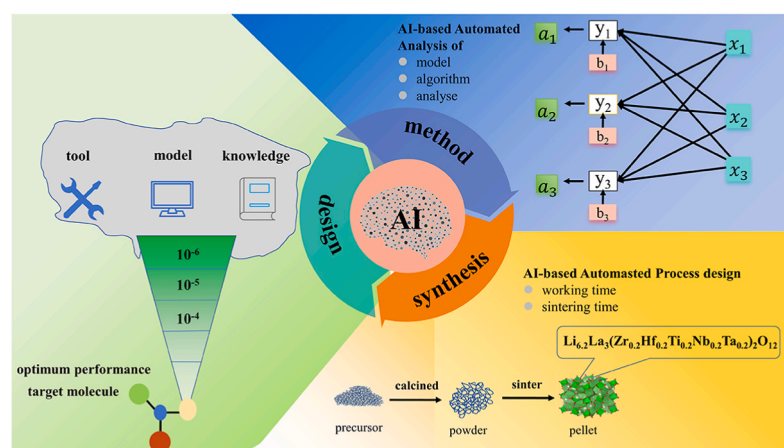
^a College of Materials Science and Engineering, Taiyuan University of Technology, Taiyuan, 030024, China

^b Shanxi Joint Laboratory of Coal based Solid Waste Resource Utilization and Green Ecological Development, Taiyuan, 030024, China

HIGHLIGHTS

- High accuracy of prediction of LLZO ionic conductivity.
- The optimized GBR model predicts the LLZO ionic conductivity.
- The feature parameters related to ionic conductivity are given.
- A database of ionic conductivity for garnet-type SSEs was established.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Solid-state batteries
Ionic conductivity
Machine learning
Garnet electrolytes
GBR optimization

ABSTRACT

Garnet solid-state electrolytes have become one of the most promising electrolyte materials due to their high ionic conductivity, wide electrochemical window, and excellent electrochemical stability. However, the trial-and-error method used to screen high-performance garnet solid-state electrolytes has the disadvantages of a long development cycle and high cost. Machine learning methods based on big data can be independent of the physical mechanism of the material, and improved the efficiency of material development. In this work, the effect of structural factor (t), first ionization energy, and other feature descriptors on ionic conductivity were studied by using the Gradient boosting regression (GBR), Random Forest (RF), eXtreme Gradient Boosting (XGB), and other models. Machine learning models can improve the accuracy of predicting the ionic conductivity of garnet solid-state electrolytes, and have guided the preparation of five garnet-type solid electrolyte materials, the ionic conductivity of $\text{Li}_{6.2}\text{La}_3\text{Zr}_{0.25}\text{Fe}_{0.25}\text{O}_{12}$ is $1.08 \times 10^{-4} \text{ S cm}^{-1}$, while the predicted value is $1.37 \times 10^{-4} \text{ S cm}^{-1}$, the descriptors in the model will provide a reference for other researchers.

* Corresponding author. College of Materials Science and Engineering, Taiyuan University of Technology, Taiyuan, 030024, China.

E-mail address: wangyongzhen@tyut.edu.cn (Y. Wang).

<https://doi.org/10.1016/j.jpowsour.2024.234492>

Received 12 February 2024; Received in revised form 22 March 2024; Accepted 1 April 2024

Available online 8 April 2024

0378-7753/© 2024 Elsevier B.V. All rights reserved.

1. Introduction

Lithium batteries are widely used in electronic devices because of their high energy density and long cycle life. However, traditional lithium-ion batteries have low safety due to their ignitability [1]. While solid-state batteries have increasing energy density [2], thinner volume size, and high safety [3] compared with traditional lithium-ion batteries, more likely to replace lithium-ion batteries [4]. Several types of solid-state electrolytes have been found such as NASICON [5–7], LISICON [8], garnet [9], sulfide [10], and perovskite-type [11,12] electrolytes, especially, $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) a garnet-type solid-state electrolyte, has wide electrochemical window [13], high ionic conductivity [14], and good compatibility with lithium [15]. However, the current ionic conductivity of LLZO cannot reach the level of liquid electrolyte, and improving the ionic conductivity is still the focus of research [16]. Therefore, researchers have focused on controlling garnet component content to enhance ionic conductivity. The traditional modified technology of LLZO is element substitution or doping, but only using vicarious trial and error search for excellent performance, much time is wasted in synthesis and measurement.

Using machine learning methods to collect data and construct relevant theoretical models can greatly reduce development cycles, reduce costs, and facilitate the prediction of high-performance materials in the model [17,18]. A model of ionic conductivity was constructed by Sendek et al. [19], which reduces the list of candidate materials from 12,831 down to 21 structures that show promise as electrolytes, few of which have been examined experimentally. Choi et al. [20] collected 14,238 elasticity information from 17,619 solid-state electrolytes, and a machine learning regression algorithm was implemented to screen for mechanically superior solid-state electrolytes. Jo et al. [21] used a machine learning-based regression model to predict the mechanical properties of Na-SSEs and collected 12,361 materials from the Materials Project.

In this work, as shown in Fig. 1, a garnet solid electrolyte database was constructed using machine learning by combining material design, methods, and synthesis. Multiple models were used to identify various factors affecting garnet solid-state electrolyte performance. The best prediction model is GBR, the correlation coefficient (R^2) of training and testing reaches 0.9. According to GBR model, we prepared and tested some garnet solid-state electrolyte, ionic conductivity of garnet solid-

state electrolytes predicted and experiment are similar. The ionic conductivity of $\text{Li}_{6.2}\text{La}_3\text{Zr}_2\text{Fe}_{0.25}\text{O}_{12}$ (LLZFeO) is $1.37 \times 10^{-4} \text{ S cm}^{-1}$, and predicted ionic conductivity is $1.08 \times 10^{-4} \text{ S cm}^{-1}$; the ionic conductivity of $\text{Li}_{6.2}\text{La}_3(\text{Hf}_{0.2}\text{Zr}_{0.2}\text{Ti}_{0.2}\text{Nb}_{0.2}\text{Ta}_{0.2})_2\text{O}_{12}$ (HEG) is $1.42 \times 10^{-4} \text{ S cm}^{-1}$, and predicted ionic conductivity is $2.62 \times 10^{-4} \text{ S cm}^{-1}$. The calculation of ionic conductivity by Density Functional Theory (DFT) is time-consuming. In summary, quick design and precise guidance for new materials synthesis can greatly simplify the time of researchers, this method gives researchers a lot of time to find out the mechanism. A combination of theoretical calculation and experiment will further highlight the advantages of machine learning.

2. Method

2.1. Database construction

398 pieces of the data were collected from open-source databases such as Materials Project (MP) [22], the Open Quantum Materials Database (OQMD) [23], and Automatic-FLOW for materials discovery (AFLOW) [24]. In addition, the corresponding feature descriptors [25] such as electronegativity, tolerance factor, ionization energy, affinity energy, and ionic radius are also set up according to the available database. After the database was established, 398 solid-state electrolyte materials with different chemical compositions were gathered, and the connection between the ionic conductivity of garnet solid-state electrolytes and various structural parameters has explored, by using decision tree (DT) [26], K-nearest neighbor (KNN) [27], Least absolute shrinkage and selection operator (LASSO) [28], Ridge regression (RR) [29] and other models. Database consists of two parts, one is the inherent properties of the material, such as atomic mass, density, electronegativity, ionization energy, and so on. These parameters are used as descriptors to predict performance. The other is the performance parameters of garnet solid electrolyte, such as ionic conductivity, activation energy, and electrochemical window.

In the database, Feature descriptors are very important [30], such as ionic radius [31], electronegativity [32], and first ionization energy [33]. These inherent properties significantly affect the ionic conductivity in the garnet solid electrolyte. In addition, the ionic conductivity in the data set is not distinguishable of bulk and total values, as these two are not always separated [34], only a few literatures make this distinction.

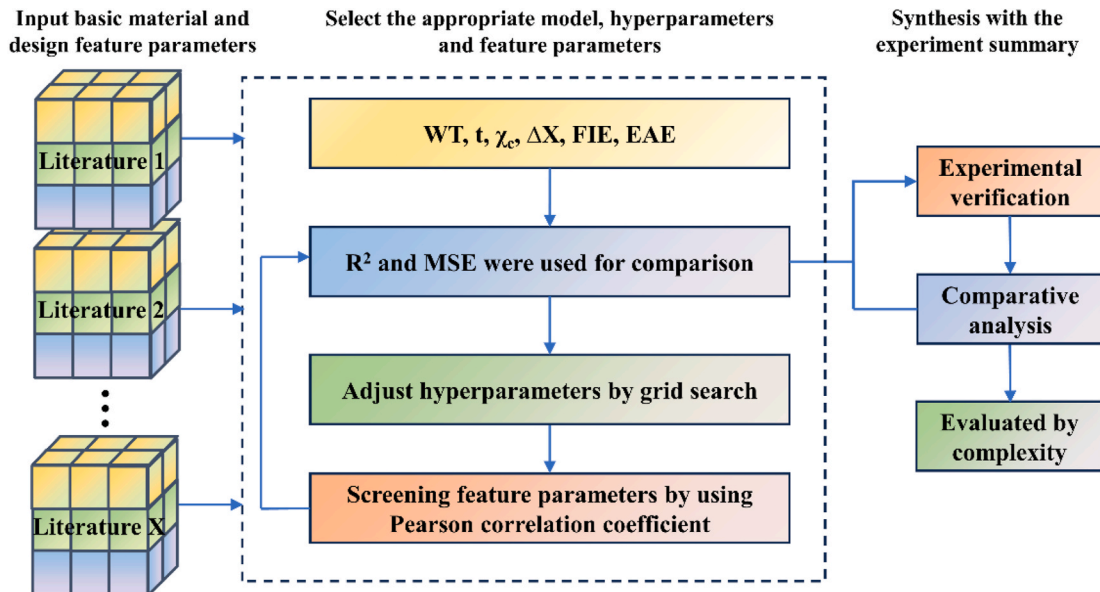


Fig. 1. The process of using machine learning for prediction. Collect data, use machine learning methods to analyze models, synthesize materials, and design new materials.

In terms of data investigation, the literature on garnet solid electrolytes was gathered, and the related properties of materials in the literature were extracted to ensure that there was clear data on their properties. Relevant performance data such as ionic conductivity, activation energy [35], preparation temperature, working temperature, and other parameters were added to the database.

2.2. Build descriptor

Garnet materials also currently have a tolerance factor to calculate the stability of the material, we use the garnet material tolerance factor t as a parameter to measure the stability of the structure [36]. Octahedron [37] and tetrahedron are distributed in the dodecahedron, which can make the material stable and also help to understand the crystal chemistry of the garnet compound. The tolerance factor is very important in materials science.

For garnet-type materials, large ionic radii such as La^{3+} will be in the large polyhedral position (dodecahedron), Li^+ and other small ionic radii will be in the tetrahedral position, and others such as Zr^{4+} , and Ta^{5+} are in the octahedral position [38], R_A , R_B , R_C , and R_D represent tetrahedrons, dodecahedrons, octahedrons, and the ionic radius of the connection position respectively. For the convenience of calculation, the dodecahedron is simplified into a cube, to facilitate calculation, the dodecahedron is simplified into a cube, and the relationship between its side lengths and diagonal lines L is as follows Eq. (1):

$$3L = 2\sqrt{3}(R_B + R_D) \quad (1)$$

The distance between the octahedron and the dodecahedron is related to L , where the center of the regular triangle formed by L as the side and the ion at point C form a right triangle, the formula is as following Eqs. (2)–(4):

$$\sqrt{3}L' = 2\sqrt{3(R_C + R_D)^2 - L^2} \quad (2)$$

H is the height of a regular tetrahedron with sides of L' :

$$\sqrt{3}H = \sqrt{2}L' \quad (3)$$

$$\sqrt{3}L' = 2\sqrt{2}(R_A + R_D) \quad (4)$$

The tolerance coefficient t is the ratio of the distance L' to the height H of the tetrahedron, as following Eq. (5):

$$t = \frac{L'}{H} = \frac{3\sqrt{(R_C + R_D)^2 - \frac{4}{9}(R_B + R_D)^2}}{2(R_A + R_D)} \quad (5)$$

The actual meaning of this tolerance factor t is the distance distribution of the octahedron and tetrahedron in the dodecahedron. When the value of t is near 1, it proves that the structure is relatively stable.

The first ionization energy is the minimum energy required to remove an electron in an atom from its ground state (the lowest energy state). It is commonly used to describe the ability of an atom or molecule to release electrons to form positive ions. With more inner electrons in the atom, the first ionization energy will be lower, Metals change from atoms to ions easily, and ions with a large charge move more easily in an electric field, so they usually have higher electrical conductivity.

Electronegativity is a scale of the ability of the atoms of an element to attract electrons in the compound [39]. The greater an element's electronegativity, the stronger its atoms' ability to attract electrons in the compound. The difference in electronegativity will affect the migration rate of ions. This will lead to the imbalance of the number of positive and negative ions in the electrolyte, which will affect the migration rate of ions, the electronegativity difference will affect the migration rate of ions. In general, ionic conductivity is proportional to the rate of ionic migration. More electronegative ions generally have greater ionic mobility because they are more likely to attract and release surrounding solvent molecules. Differences in electronegativity can also affect the

crystal structure and thus ions' arrangement and migration paths.

In some cases, ions with higher electronegativity may fill vacancies or channels in the crystal, which can provide more ionic migration paths and thus increase ionic conductivity. The actual descriptors used are shown in Table 1. Specific descriptor selections and adjustments are placed in Table S1.

2.3. Prediction model

A database of 398 pieces of data was created, containing parameters such as material element composition, ionic conductivity, electronegativity, atomic mass, and first ionization energy. The ionic conductivity ranges from 8.7×10^{-9} to $7.08 \times 10^{-2} \text{ S cm}^{-1}$, with log 10 values ranging from -8.06 to -1.15 , with an average of -4.13 and a median of -3.91 in Fig. 2(a).

While building the database, the material structures were manually classified, all of the materials were garnet solid-state electrolytes. More specific ion conductivity distribution can be found in Figs. S1–S3.

In this paper, a machine learning model (GBR) is presented. 398 solid-state electrolyte data are input to obtain corresponding material data and ionic conductivity descriptors are designed. GBR model has the advantages of high accuracy, great robustness and Strong interpretability: High accuracy, each algorithm learns from the mistakes of the previous learning algorithm, it has the role of iterative learning; Great robustness, the combination of multiple algorithms can effectively reduce the influence of outliers in the data; Strong interpretability, the importance of the features of each algorithm and the learning process of the model are clear. The data are divided into 10 parts by data set partitioning method using the scikit-learn library, among which 8 parts are used as the training set and 2 part is used as the test set. The K-fold cross-validation method is also adopted to carry out 10-fold cross-validation, which also achieves good results. The hyperparameters are set to learning_rate = 0.001, max_depth = 11, min_samples_leaf = 1, min_samples_split = 2, and n_estimators = 300.

Random forest (RF) [40] is a kind of learning machine based on a decision tree, which can complete the learning task by constructing and combining multiple weak learners. At the same time, it can resist overfitting and is insensitive and sensitive to missing data [41], so it is suitable for high-dimensional and unbalanced sample data [42]. Among the features proposed above, 10 feature descriptors were selected for the model, such as ionic conductivity, electronegativity, tolerance factor t , and first ionization energy. The remaining eigenvalues, such as preparation temperature, activation energy, and atomic mass were rejected due to unimportant data.

When the feature parameters are highly correlated, such as atomic mass variance and atomic mass standard deviation, only one parameter is needed to express the other parameters. When there are too many such parameters, it will affect the decision of the model, and it will also be the feature parameter redundancy. Therefore, we used the Pearson correlation coefficient [43] in statistics to measure the correlation of each

Table 1

The 10 descriptors used for prediction in the model are judged by feature importance and correlation matrix.

Characteristic	Description	Type	Range
total C	Element occupancy at point C	float	[0,1]
WT	work temperature	int	[0,+∞)
R_C Var	The variance of the radius of R_C	float	(-∞,+∞)
FIE AVG C	The average first ionization energy of point C	float	[0,+∞)
EAE AVG C	The average Electron affinity energy of point C	float	[0,+∞)
EAE SD C	The standard deviation of Electron affinity energy C	float	(-∞,+∞)
t	Structure factor, the ratio of L' to H	float	[0,+∞)
χ_C	The average electronegativity of point C	float	[0,+∞)
χ_C Var	the Variance of electronegativity C	float	(-∞,+∞)
$\Delta S R_C$	Configurational entropy of R_C	float	(-∞,+∞)

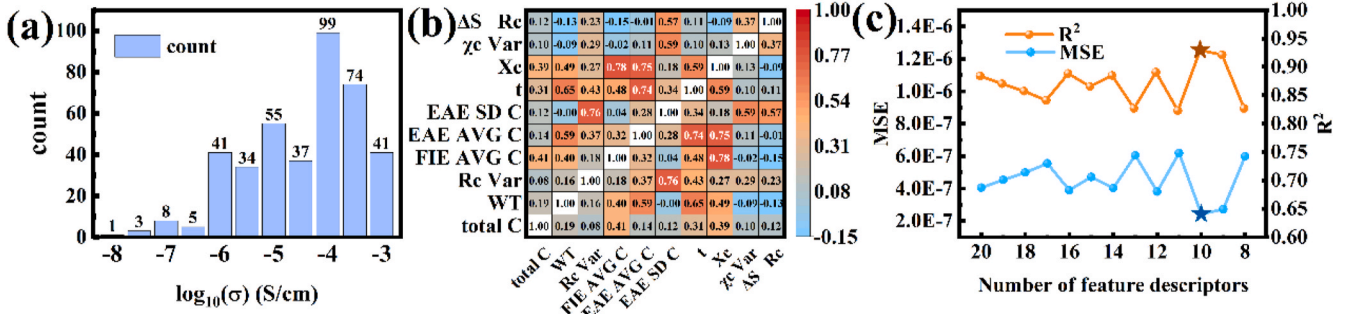


Fig. 2. (a) Distribution conductivities for materials in the database. A histogram displaying the 398 ionic conductivities from materials contained in this database. (b) The correlation matrix is displayed, and r has little correlation in each feature parameter, indicating that the correlation between features is weak enough to be represented separately. (c) The influence of the feature numbers on R^2 and MSE in GBR model.

feature parameter, Pearson correlation coefficient formula is as follows Eq. (6):

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (6)$$

Where r represents the Pearson correlation coefficient, X_i , Y_i represents the value of the feature parameter in the sample $\bar{X}$, $\bar{Y}$ represents the mean value of the feature parameter, and n represents the number of data points. To reduce the correlation of feature parameters, we deleted the high positive correlation (>0.8) and a high negative correlation (<-0.8) of the characteristic parameters, the above characteristic parameters are obtained, and the following is a thermal map using Pearson correlation coefficients, as shown in Fig. 2(b). At the same time, we adjusted the number of feature descriptors, as shown in Fig. 2(c). Only 10 feature descriptors are selected for the model, and GBR model has high R^2 and low MSE, indicating that the use of model prediction has a good effect in this case. MSE is a metric to measure the accuracy of a data prediction model. The smaller the MSE, the lower the deviation and the higher the accuracy.

We used the Python library “train test split” as a dataset partitioning method. Under this processing, 80% of the data is divided into the training set and the remaining 20% is used as the test set [44]. Each training is on a slightly different training set, resulting in different results. “train-test-split” provides a more efficient result than the training test alone. “ r^2 _score” of the “scikit-learn” package is used to judge the model fitting effect, and “mean-square error” model is used to judge the mean square error. As shown in Fig. 3, When RF, XGB, and GBR were

used to predict the model, the accuracy of the training set was 0.83, 0.89, and 0.90, the accuracy of the test set was 0.84, 0.89, and 0.90, respectively, and the mean square error (MSE) was 1.15×10^{-4} , 3.38×10^{-4} and 1.57×10^{-7} .

The accuracy of other models, such as Linear Ridge Regression (LRR) [45], Orthogonal Matching Pursuit (OMP) [46], and Least Angle Regression (LAR) [47], is 0.62, 0.62, and 0.63 respectively, which is a little worse than the random forest effect. Based on this situation, GBR model has the highest R^2 and lowest MSE, we choose the GBR model as the algorithm to predict the ionic conductivity of garnet solid electrolyte, normalize the model, draw the interactive residuals, and analyze the distribution of material data and the existence of outliers.

After drawing the Normal QQ-Plot [48], as shown in Fig. 4(a), we find that the points in the plot roughly fall on a straight line, indicating that the data approximately follows a normal distribution. However, some data points are still not on the line, and these outliers may be data outliers. As shown in Fig. 4(b), the Tukey-Anscombe plot is a visual tool used to evaluate the suitability of a regression model by comparing the statistical characteristics of multiple data sets to check how well the regression model fits. When the predicted value of the predictor and the actual observed value are close to a straight line, it indicates that the fitting effect is good.

2.4. Shapley Additive exPlanations (SHAP)

According to the model formula, the feature importance [49] graph is derived. The higher the value, the greater the influence of the feature parameter on the target performance. There are many algorithms to calculate the importance of features [50], such as the calculation based on the number of features used in the model and the average degree of

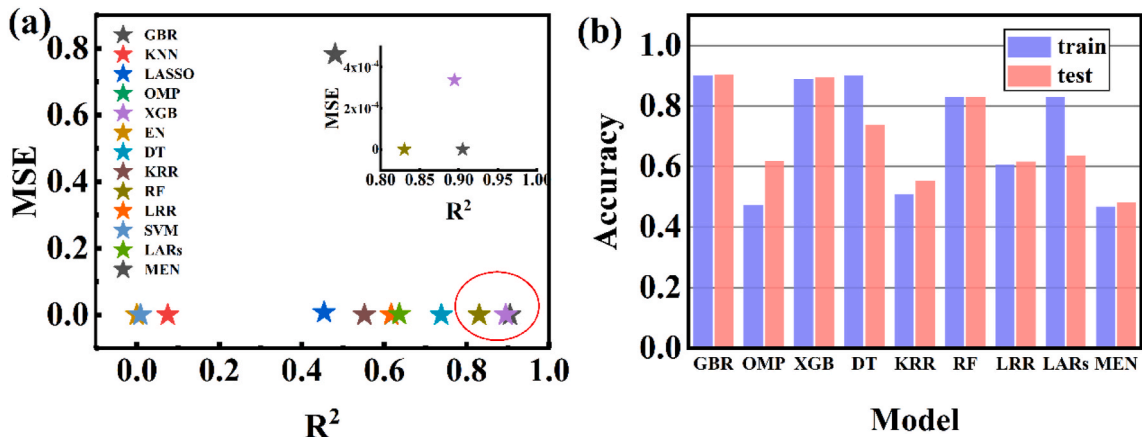


Fig. 3. The following are the fitting effects R^2 and mean square error (MSE) of the training model, (a) the results of all training models are shown, and (b) Accuracy in training set and test set for different ML models.

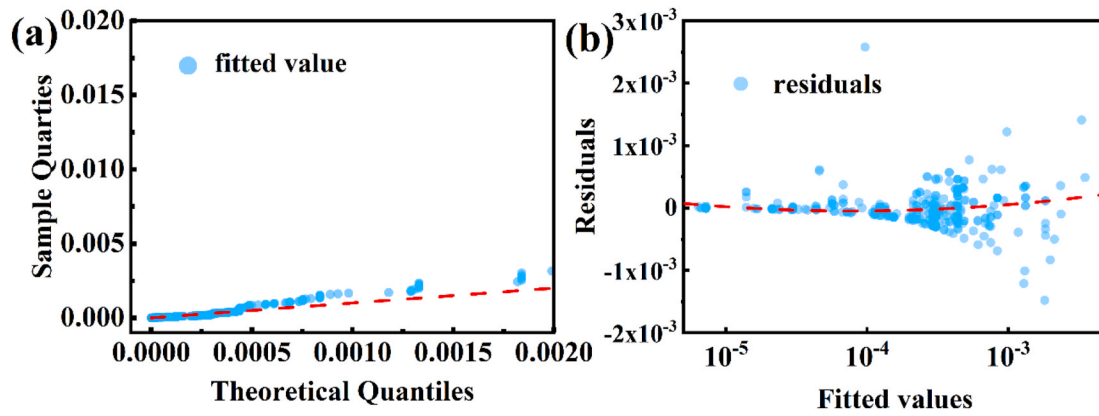


Fig. 4. (a) Normal QQ-Plot is a visual tool for verifying that the data conform to a normal distribution. (b) Tukey-Anscombe plot is a visual tool used to evaluate the applicability of regression models.

reducing the degree of impurity; the value of a feature can also be randomly disturbed to observe the change of the model's predictive performance; the greater the absolute value of the coefficient of the linear model, the greater the influence of the feature parameter on the target variable; or construct multiple training trees for each feature parameter, count the number of times each feature parameter is used, and calculate its proportion as a measure of feature importance. We chose the last method to calculate the importance of features, which has the advantage of accurately understanding the number of features used, reducing model complexity and calculation cost, and improving model generalization ability and efficiency. The importance of specific features is shown in Fig. 5(a).

The shap diagram explains the feature parameter prediction method by calculating the contribution of each feature parameter to the target value [51]. It can provide information about the influence of specific feature parameters on the target value, and help understand how the model uses the information of different features to predict. Through color coding and the position of the features, the positive or negative influence of the features can be displayed. As shown in Fig. 5(b), the increase in the proportion of elements at point C (total C) is not conducive to the improvement of ionic conductivity. By comprehensive analysis, total C indicates that the proportion of Zr site (C point) in the compound decreases, and Zr atoms need to be replaced by high-priced elements to improve the ionic conductivity. Doped elements such as Ta^{5+} and Nb^{5+} will likely produce the effect of high-priced ions and increase the ionic conductivity. Similarly, after hypervalent ions are mixed, the ionic radius will also be increased, which is consistent with the positive correlation between standard deviation (EAE SD C) and

ionic conductivity at point C. Ionic conductivity is negatively correlated with electronegativity (χ_c), ionization energy (FIE AVG C), and tolerance factor t which indicates that when doping high hypervalent ions at point C, electronegativity needs to be kept within a reasonable range, and ionization energy also needs to be kept at a low level to enable atoms to become ions, FIE AVG C, χ_c , EAE AVG C all show that the ionic conductivity of replaceable elements will increase when they are at the bottom left of the periodic table. In garnet solid-state electrolytes, octahedral structure is the key structure connecting tetrahedrons and dodecahedrons. The reduction of its ionic radius will make the garnet materials more stable, and the reduction of tolerance factor t will improve the structural stability. The remaining characteristic parameters, such as χ_c Var, R_c Var, and EAE SD C, all increase the ionic conductivity with the parameter increase, and these parameters represent the element complexity at the Zr site, indicating that replacing other elements at the Zr site will help improve the ionic conductivity. In summary, it can be concluded that high-priced elements at the bottom left of the periodic table are more likely to increase the ionic conductivity of the LLZO system.

Cook distance [52] is a statistic used to detect outliers or outliers in a statistical model, which can be used to assess the extent to which a data point affects the overall fit of a statistical model. It is measured by calculating how the model parameters change after removing the data point. When a data point greatly impacts the model's overall fitting effect, its Cook distance value will be larger. This is also consistent with the previous outlier detection results. Outliers (cook's distance >0.5) affect the prediction results and reduce the fitting effect. According to professional advice, we deleted the corresponding data to prevent a

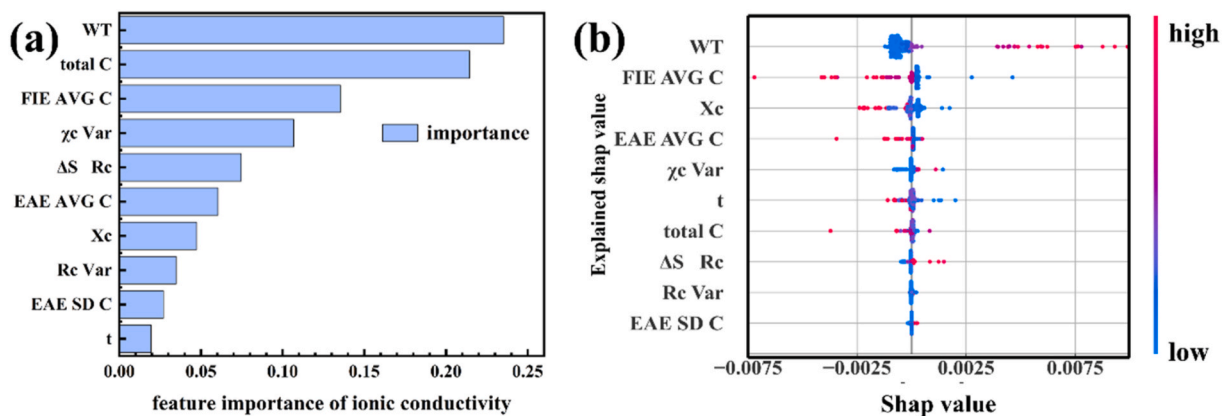


Fig. 5. (a) Feature importance of ionic conductivity in garnet solid-state electrolyte. The greater the value of feature importance in the graph, the more times it is used in the model. (b) Interpretability analysis of shap graphs. The influence of different characteristic parameters on the ionic conductivity of the material and the relevant influence degree are shown in shap.

negative impact on the model fitting effect. As shown in Fig. 6.

After finding outliers, we discussed with relevant researchers, combined with relevant database information, and conducted correctness tests on samples in turn based on data and expert experience. Abnormal data was found as follows in Table 2.

After analysis, the results are obtained, the 1st material contains aluminum, the presence of aluminum will improve the ionic conductivity, and the ionic conductivity of the material is too low, and does not meet the actual situation. The 2nd material has too high ionic conductivity, the structural composition is conventional, and the actual conductivity should be about 10^{-4} S cm^{-1} . The rest materials are conventional garnet materials, adding one or two elements, ionic conductivity is too low. In practice, doped Ta, Nb garnet materials should significantly improve the ionic conductivity [53], so it is listed as an outlier point.

3. Result and discussion

We used the model to calculate 90 feature descriptors and selected several parameters with the highest feature importance. Most of the useless features have been removed, and the remaining feature parameters have an effect on the ionic conductivity and have a certain impact on the performance. In addition, as shown in Fig. 7, we used a large number of models to fit the data, looking for a model suitable for predicting ionic conductivity, with a high fitting degree and low deviation.

From the analysis of Fig. 5(a), the Zr site (total C) plays an important role in garnet-type materials. Such as element occupancy rate of the Zr site, ionization energy, and electronegativity, these feature descriptors make a great contribution to the model fitting. In high-entropy materials, multi-element equal molar ratio doping will change the performance of the material, such as the “cocktail effect”, lattice distortion effect, etc. In the data collection, it is inevitable to collect a lot of garnet solid-state electrolytes with high-entropy materials. Various elements are often doped at the Zr site for this high-entropy material. Therefore, the Zr site plays an important role in the prediction of high-entropy garnet-type materials.

It can be seen from the Fig. 5(b) that “total C” has a positive correlation with ionic conductivity, because the doped high plateau number element forms a higher ionic radius and charge, which is convenient to promote Li^+ migration, but the higher occupancy rate will reduce the occupancy rate of other elements. The octahedral structure of the C site also plays an important role in the lattice. The change of the tolerance factor t will make the structure unstable, so the occupancy rate of the

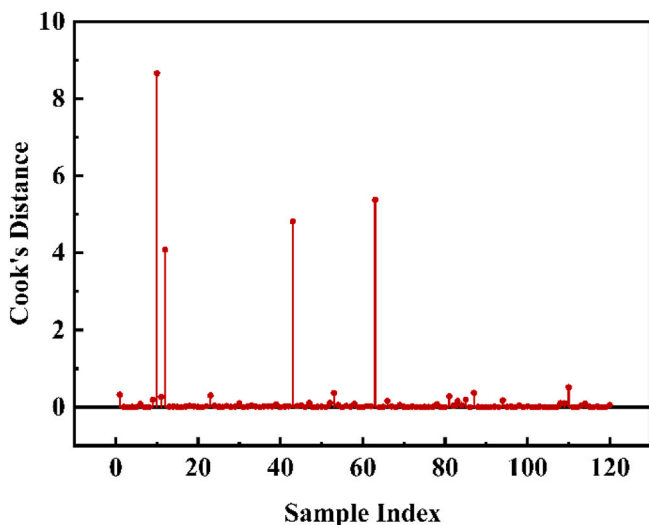


Fig. 6. Abnormal values and cook's distance in samples. The farther the point is from the horizontal line, the more obvious the outlier.

Table 2

5 outliers selected from the original material collection.

composition	Ionic conductivity S cm^{-1}	reason
$\text{Li}_7\text{La}_{2.9}\text{Al}_{0.1}\text{Zr}_2\text{O}_{12}$	4.1×10^{-6}	The actual ionic conductivity can be improved
$\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.5}\text{Nb}_{0.1}\text{O}_{12}$	1.5×10^{-3}	The actual ionic conductivity should be 10^{-4}
$\text{Li}_5\text{BaLa}_2\text{Ta}_2\text{O}_{11.5}$	4.9×10^{-6}	The actual ionic conductivity can be improved
$\text{Li}_{5.4}\text{La}_3\text{Nb}_{1.8}\text{Sm}_{0.2}\text{O}_{12}$	3.71×10^{-5}	The actual ionic conductivity can be improved

point Zr element is very important. Similarly, the ionization energy and electron affinity energy are also important to ionic conductivity, the transformation of atoms into ions will improve the ion conduction ability, while preventing the thermal expansion and phase transition of the lattice, improving the stability and high-temperature resistance. Stability parameter t indicates the degree of stability of dodecahedron and octahedron, through the calculation of the radius of each ion, the tolerance coefficient of garnet material can be determined. The garnet structure is more stable when the tolerance coefficient is about 1.

After considering the above conditions, we prepared several garnet-type materials. Ionic conductivity of $\text{Li}_{6.2}\text{La}_3(\text{Zr}_{0.2}\text{Hf}_{0.2}\text{Ti}_{0.2}\text{Nb}_{0.2}\text{Ta}_{0.2})_2\text{O}_{12}$ (HEG) [53], $\text{Li}_{7.4}\text{La}_3\text{Zr}_{1.6}\text{In}_{0.4}\text{O}_{12}$ (LLZInO), $\text{Li}_7\text{La}_3\text{Zr}_{1.6}\text{Ce}_{0.4}\text{O}_{12}$ (LLZCeO), $\text{Li}_{6.2}\text{La}_3\text{Zr}_2\text{Fe}_{0.25}\text{O}_{12}$ (LLZFeO), $\text{Li}_{6.2}\text{La}_3\text{Zr}_2\text{Fe}_{0.25}\text{O}_{12}$ (LLZFeO) are 1.42×10^{-4} , 2.66×10^{-5} , 1.51×10^{-4} , 1.37×10^{-4} , and 4.06×10^{-4} S cm^{-1} , respectively, and predicted ionic conductivity are 2.62×10^{-4} , 6.70×10^{-5} , 2.47×10^{-4} , 1.08×10^{-4} , and 5.07×10^{-4} S cm^{-1} . The actual conductivity of these materials is similar to the predicted conductivity, as shown in Fig. 8. We believe that using this model will significantly reduce material development time and improve efficiency.

4. Conclusion

In this study, a GBR model trained with 398 solid-state electrolytes showed R^2 values of 0.90 and 0.90 for the train score and test score (train set: 0.8, train set: 0.2). Solid-state electrolytes doped with different elements are prepared by GBR model screening, five garnet solid-state electrolytes were synthesized, the predicted conductivity of materials is close to the actual conductivity, shown that the GBR model has a high accuracy in predicting the ionic conductivity of garnet-type materials, compared with DFT, it is more rapid and reduces a lot of time. Feature descriptor such as work temperature, first ionization energy, and tolerance factor t play an important role in predicted ionic conductivity. In the preparation of high ionic conductivity garnet solid-state electrolyte, the above 10 feature descriptors can be considered.

CRedit authorship contribution statement

Yue Ma: Writing – original draft, Methodology. **Shaoxiong Han:** Resources, Investigation. **Yan Sun:** Resources, Investigation, Data curation. **Zhenming Cui:** Resources, Investigation, Data curation. **Pengyu Liu:** Resources, Investigation, Data curation. **Xiaomin Wang:** Investigation, Data curation. **Yongzhen Wang:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

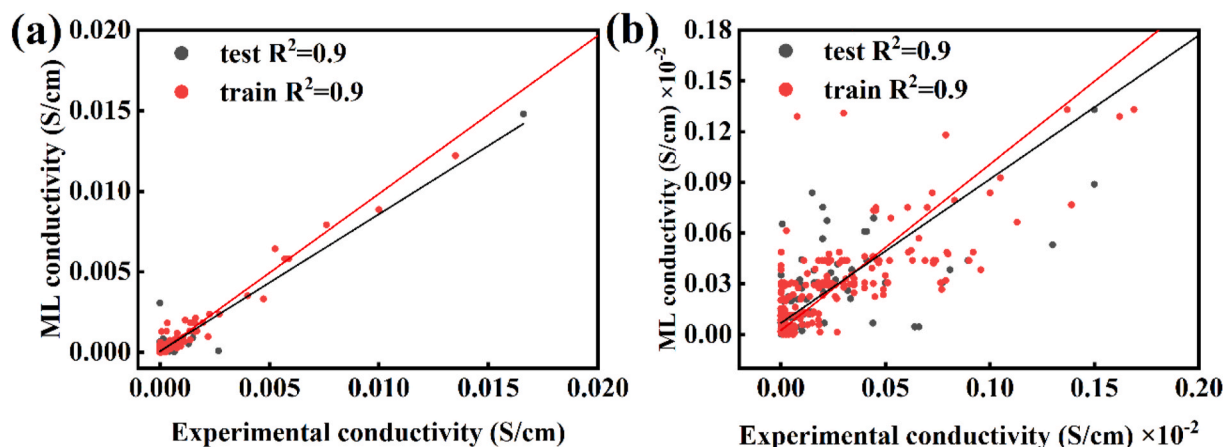


Fig. 7. (a) Fit plots for testing and verifying training data. (b) Showing partially enlarged images. The fitting effect R^2 of predicted data and actual data was 0.9.

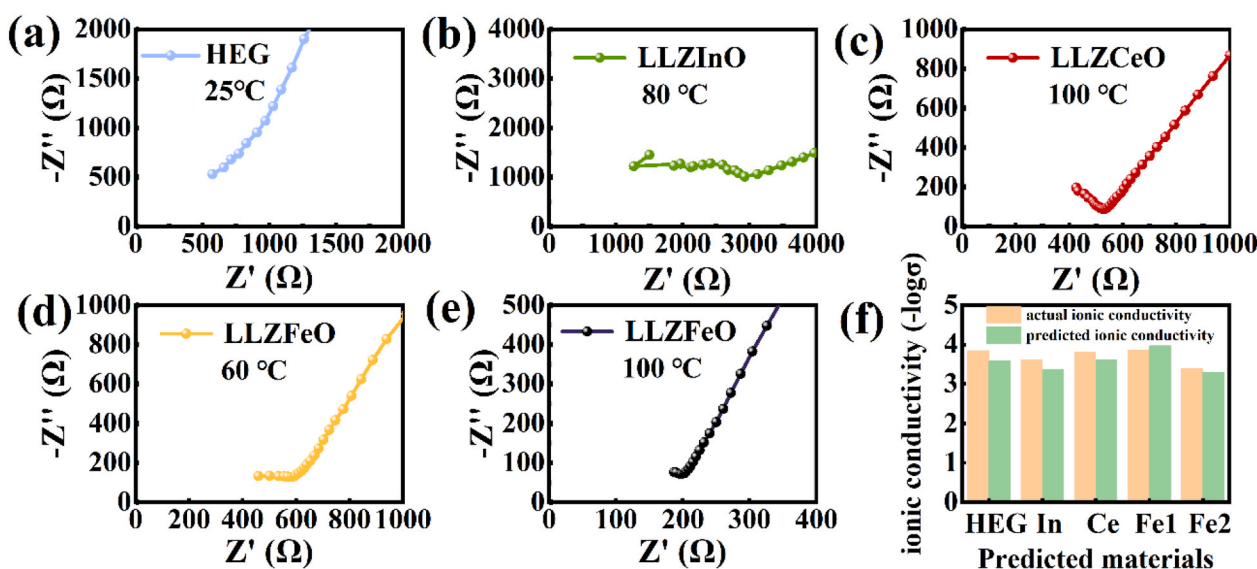


Fig. 8. (a) Nyquist plots of HEG pellets measured at 25 °C where Z' refers to the real impedance part of the AC impedance response of the electrochemical system related to ion transport, which usually corresponds to the diffusion process of ions in the electrolyte, and $-Z''$ refers to the imaginary impedance part related to the electrode surface reaction in the AC impedance response of the electrochemical system, which usually corresponds to the electron transfer process. (b) Nyquist plots of LLZInO pellet measured at 80 °C. (c) Nyquist plots of LLZCeO pellet measured at 100 °C. (d) Nyquist plots of LLZFeO pellet measured at 60 °C. (e) Nyquist plots of LLZFeO pellet measured at 100 °C. (f) Predicted ionic conductivity and actual ionic conductivity of five prepared materials.

Data availability

Data will be made available on request.

Acknowledgments

This work was funded by the National Natural Science Foundation of China (Grant No. 52371231), the Key Research and Development Program of Shanxi Province (Grant No. 202102030201006), and the Central Government Guides Local Science, Technology Development Special Fund Project (Grant No. YDZJSX2022B003), and Key R&D program of Shanxi Province (Grant No. 202302040201008).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2024.234492>.

References

- [1] P. Yao, H. Yu, Z. Ding, Y. Liu, J. Lu, M. Lavorgna, J. Wu, X. Liu, Review on polymer-based composite electrolytes for lithium batteries, *Front. Chem.* 7 (2019) 522.
- [2] X. Wu, X. Liang, X. Zhang, L. Lan, S. Li, Q. Gai, Structural evolution of plasma sprayed amorphous $\text{Li}_4\text{Ti}_5\text{O}_{12}$ electrode and ceramic/polymer composite electrolyte during electrochemical cycle of quasi-solid-state lithium battery, *J. Adv. Ceram.* 10 (2021) 347–354.
- [3] J.G. Kim, B. Son, S. Mukherjee, N. Schuppert, A. Bates, O. Kwon, M.J. Choi, H. Y. Chung, S. Park, A review of lithium and non-lithium based solid state batteries, *J. Power Sources* 282 (2015) 299–322.
- [4] J. Sun, S. Kang, J. Kim, K. Min, Accelerated discovery of novel garnet-type solid-state electrolyte candidates via machine learning, *ACS Appl. Mater. Interfaces* 15 (2023) 5049–5057.
- [5] H. Tian, S. Liu, L. Deng, L. Wang, L. Dai, New-type HF-based NASICON electrolyte for solid-state Na-ion batteries with superior long-cycling stability and rate capability, *Energy Storage Mater.* 39 (2021) 232–238.
- [6] J. Gao, J. Wu, S. Han, J. Zhang, L. Zhu, Y. Wu, J. Zhao, W. Tang, A novel solid electrolyte formed by NASICON-type $\text{Li}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ and poly(vinylidene fluoride) for solid state batteries, *Funct. Mater. Lett.* 14 (2021) 2140001.
- [7] W. Go, I.-S. Jang, B.-Y. Song, H. Park, Y.C. Kang, J. Chun, Improving ionic conductivity of von-Alpen-type NASICON ceramic electrolytes via magnesium doping, *J. Adv. Ceram.* 12 (2023) 1058–1066.

- [8] B. Zhang, L. Yang, L.-W. Wang, F. Pan, Cooperative transport enabling fast Li-ion diffusion in Thio-LISICON $\text{Li}_{10}\text{SiP}_2\text{S}_{12}$ solid electrolyte, *Nano Energy* 62 (2019) 844–852.
- [9] S. Han, Z. Wang, Y. Ma, Y. Zhang, Y. Wang, X. Wang, Recent advances in solving Li_2CO_3 problems in garnet-based solid-state battery: a systematic review (2020–2023), *J. Energy Chem.* 90 (2023) 58–76.
- [10] K.J. Kim, M. Balaish, M. Wadaguchi, L. Kong, J.L.M. Rupp, Solid-state Li–metal batteries: challenges and horizons of oxide and sulfide solid electrolytes and their interfaces, *Adv. Energy Mater.* 11 (2020) 2002689.
- [11] Y. Li, H. Xu, P.H. Chien, N. Wu, S. Xin, L. Xue, K. Park, Y.Y. Hu, J.B. Goodenough, A perovskite electrolyte that is stable in moist air for lithium-ion batteries, *Angew. Chem. Int. Ed.* 57 (2018) 8587–8591.
- [12] Y. Jia, S. Chen, X. Shao, J. Chen, D.-L. Fang, S. Li, A. Mao, C. Li, Synergetic effect of lattice distortion and oxygen vacancies on high-rate lithium-ion storage in high-entropy perovskite oxides, *J. Adv. Ceram.* 12 (2023) 1214–1227.
- [13] Y. Jiang, X. Zhu, W. Lai, Three electrodes analysis of a 3 V-class all-solid-state lithium-ion battery based on garnet-type solid electrolyte $\text{Li}_{6.4}\text{La}_3\text{Zr}_{1.4}\text{Ta}_{0.6}\text{O}_{12}$, *J. Power Sources* 529 (2022) 231287.
- [14] A.K. Mishra, S. Rajput, N. Karamta, I. Mukhopadhyay, Exploring the possibility of machine learning for predicting ionic conductivity of solid-state electrolytes, *ACS Omega* 8 (2023) 16419–16427.
- [15] Z.-h. Zhang, T. Wei, J.-h. Lu, Q.-m. Xiong, Y.-h. Ji, Z.-y. Zhu, L.-t. Zhang, Practical development and challenges of garnet-structured $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ electrolytes for all-solid-state lithium-ion batteries: a review, *Int. J. Miner. Metall. Mater.* 28 (2021) 1565–1583.
- [16] C. Zhang, X. Hu, Z. Nie, C. Wu, N. Zheng, S. Chen, Y. Yang, R. Wei, J. Yu, N. Yang, Y. Yu, W. Liu, High-performance Ta-doped $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ garnet oxides with AlN additive, *J. Adv. Ceram.* 11 (2022) 1530–1541.
- [17] Y. Qiu, Z. Wu, J. Wang, C. Zhang, H. Zhang, Introduction of materials genome technology and its applications in the field of biomedical materials, *Materials* 16 (2023) 1906.
- [18] V. Fung, P. Ganesh, B.G. Sumpter, Physically informed machine learning prediction of electronic density of states, *Chem. Mater.* 34 (2022) 4848–4855.
- [19] A.D. Sendek, Q. Yang, E.D. Cubuk, K.-A.N. Duerloo, Y. Cui, E.J. Reed, Holistic computational structure screening of more than 12 000 candidates for solid lithium-ion conductor materials, *Energy Environ. Sci.* 10 (2017) 306–320.
- [20] E. Choi, J. Jo, W. Kim, M. Min, Searching for mechanically superior solid-state electrolytes in Li-ion batteries via data-driven approaches, *ACS Appl. Mater. Interfaces* 13 (2021) 42590–42597.
- [21] J. Jo, E. Choi, M. Kim, M. Min, Machine learning-aided materials design platform for predicting the mechanical properties of Na-ion solid-state electrolytes, *ACS Appl. Energy Mater.* 4 (2021) 7862–7869.
- [22] K. Mathew, A.K. Singh, J.J. Gabriel, K. Choudhary, S.B. Sinnott, A.V. Davydov, F. Tavazza, R.G. Hennig, MPInterfaces: a Materials Project based Python tool for high-throughput computational screening of interfacial systems, *Comput. Mater. Sci.* 122 (2016) 183–190.
- [23] S. Kirklin, J.E. Saal, B. Meredig, A. Thompson, J.W. Doak, M. Aykol, S. Rühl, C. Wolverton, The Open Quantum Materials Database (OQMD): assessing the accuracy of DFT formation energies, *npj Comput. Mater.* 1 (2015) 1–15.
- [24] S. Curtarolo, W. Setyawan, S. Wang, J. Xue, K. Yang, R.H. Taylor, L.J. Nelson, G.L. W. Hart, S. Sanvito, M. Buongiorno-Nardelli, N. Mingo, O. Levy, AFLOWLIB.ORG: a distributed materials properties repository from high-throughput ab initio calculations, *Comput. Mater. Sci.* 58 (2012) 227–235.
- [25] D. Kaya, D. Koroglu, E. Aydin, B. Uralcan, Optimization of capacitance in supercapacitors by constructing an experimentally validated hybrid artificial neural networks-genetic algorithm framework, *J. Power Sources* 568 (2023) 232987.
- [26] S.B. Kotsiantis, Decision trees: a recent overview, *Artif. Intell. Rev.* 39 (2011) 261–283.
- [27] S. Zhang, X. Li, M. Zong, X. Zhu, D. Cheng, Learning k for kNN Classification, *ACM Trans. Internet Technol.* 8 (2017) 1–19.
- [28] A.S. Dalalyan, M. Hebiri, J. Lederer, On the prediction performance of the Lasso, *Bernoulli* 23 (2017) 552–581.
- [29] A.V. Dorugade, New ridge parameters for ridge regression, *J. Assoc. Arab Univ. Basic Appl. Sci.* 15 (2018) 94–99.
- [30] M.S. Beal, B.E. Hayden, T. Le Gall, C.E. Lee, X. Lu, M. Mirsaneh, C. Mormiche, D. Pasero, D.C. Smith, A. Weld, C. Yada, S. Yokoishi, High throughput methodology for synthesis, screening, and optimization of solid state lithium ion electrolytes, *ACS Comb. Sci.* 13 (2011) 375–381.
- [31] R.H. Langley, G.D. Sturgeon, Lattice parameters and ionic radii of the oxide and fluoride garnets, *J. Solid State Chem.* 30 (1979) 79–82.
- [32] M. Hosseinzadeh, A. Hemmati-Sarapardeh, F. Ameli, F. Naderi, M. Dastgahi, A computational intelligence scheme for estimating electrical conductivity of ternary mixtures containing ionic liquids, *J. Mol. Liq.* 221 (2016) 624–632.
- [33] J. Lee, N. Ohba, R. Asahi, Design rules for high oxygen-ion conductivity in garnet-type oxides, *Chem. Mater.* 32 (2020) 1358–1370.
- [34] C.J. Hargreaves, M.W. Gaultois, L.M. Daniels, E.J. Watts, V.A. Kurlin, M. Moran, Y. Dang, R. Morris, A. Morscher, K. Thompson, M.A. Wright, B.-E. Prasad, F. Blanc, C.M. Collins, C.A. Crawford, B.B. Duff, J. Evans, J. Gamon, G. Han, B.T. Leube, H. Niu, A.J. Perez, A. Robinson, O. Rogan, P.M. Sharp, E. Shoko, M. Sonni, W. J. Thomas, A. Vasylenko, L. Wang, M.J. Rosseinsky, M.S. Dyer, A database of experimentally measured lithium solid electrolyte conductivities evaluated with machine learning, *npj Comput. Mater.* 9 (2023) 9.
- [35] F.M. Pesci, A. Bertei, R.H. Brugge, S.P. Emge, A.K.O. Hekselman, L.E. Marbella, C. P. Grey, A. Agüero, Establishing ultralow activation energies for lithium transport in garnet electrolytes, *ACS Appl. Mater. Interfaces* 12 (2020) 32806–32816.
- [36] Z. Song, D. Zhou, Q. Liu, Tolerance factor and phase stability of the garnet structure, *Acta Crystallogr. C Struct. Chem.* 75 (2019) 1353–1358.
- [37] Y. Li, J.-T. Han, C.-A. Wang, H. Xie, J.B. Goodenough, Optimizing Li^+ conductivity in a garnet framework, *J. Mater. Chem.* 22 (2012) 15357–15361.
- [38] X. Yi, Y. Guo, S. Pan, Y. Wang, S. Chi, S. Wu, Q.H. Yang, Duality of Li_2CO_3 in solid-state batteries, *Trans. Tianjin Univ.* 29 (2023) 73–87.
- [39] R. Jalem, Y. Morishita, T. Okajima, H. Takeda, Y. Kondo, M. Nakayama, T. Kasuga, Experimental and first-principles DFT study on the electrochemical reactivity of garnet-type solid electrolytes with carbon, *J. Mater. Chem. A* 4 (2016) 14371–14379.
- [40] J. Schöning, H. von Eynatten, R. Tolosana-Delgado, G. Meinhold, Garnet major-element composition as an indicator of host-rock type: a machine learning approach using the random forest classifier, *Contrib. Mineral. Petrol.* 176 (2021) 1–21.
- [41] Q. Zhou, H. Zhou, T. Li, Cost-sensitive feature selection using random forest: selecting low-cost subsets of informative features, *Knowl. Base Syst.* 95 (2016) 1–11.
- [42] B.F. Darst, K.C. Malecki, C.D. Engelman, Using recursive feature elimination in random forest to account for correlated variables in high dimensional data, *BMC Genet.* 19 (2018) 65.
- [43] A. Ly, M. Marsman, E.J. Wagenmakers, Analytic posteriors for Pearson's correlation coefficient, *Stat. Neerl.* 72 (2018) 4–13.
- [44] K. Pawluszek-Filipiak, A. Borkowski, On the importance of train–test split ratio of datasets in automatic landslide detection by supervised classification, *Rem. Sens.* 12 (2020) 3054.
- [45] J.O. Ogutu, T. Schulz-Streeck, H.P. Piepho, Genomic selection using regularized linear regression models: ridge regression, lasso, elastic net and their extensions, *BMC Proc.* 6 (2012) 1–6.
- [46] O.F. Alcin, A. Sengur, J. Qian, M.C. Ince, OMP-ELM: orthogonal matching pursuit-based extreme learning machine for regression, *Int. J. Intell. Syst.* 24 (2015) 135–143.
- [47] J. Zhang, J.Y. Feng, Y.L. Ni, Y.J. Wen, Y. Niu, C.L. Tamba, C. Yue, Q. Song, Y. M. Zhang, pLARM-EB: integration of least angle regression with empirical Bayes for multilocus genome-wide association studies, *Heredity* 118 (2017) 517–524.
- [48] S.S. Dhar, B. Chakraborty, P. Chaudhuri, Comparison of multivariate distributions using quantile–quantile plots and related tests, *Bernoulli* 20 (2014) 1484–1506.
- [49] K. Kaufmann, D. Maryanovsky, W.M. Mellor, C. Zhu, A.S. Rosengarten, T. J. Harrington, C. Oses, C. Toher, S. Curtarolo, K.S. Vecchio, Discovery of high-entropy ceramics via machine learning, *npj Comput. Mater.* 6 (2020) 42.
- [50] M.S. Raza, U. Qamar, Feature selection using rough set-based direct dependency calculation by avoiding the positive region, *Int. J. Approx. Reason.* 92 (2018) 175–197.
- [51] A.B. Parsa, A. Movahedi, H. Taghipour, S. Derrbi, A.K. Mohammadian, Toward safer highways, application of XGBoost and SHAP for real-time accident detection and feature analysis, *Accid. Anal. Prev.* 136 (2020) 105405.
- [52] A.C. Rajan, A. Mishra, S. Satsangi, R. Vaish, H. Mizuseki, K.-R. Lee, A.K. Singh, Machine-learning-assisted accurate band gap predictions of functionalized MXene, *Chem. Mater.* 30 (2018) 4031–4038.
- [53] S. Han, Z. Wang, Y. Ma, Y. Miao, X. Wang, Y. Wang, Y. Wang, Fast ion-conducting high-entropy garnet solid-state electrolytes with excellent air stability, *J. Adv. Ceram.* 12 (2023) 1201–1213.